

Tabular Data Processing Pipeline (Conceptual)

1. Pre-processing

Feature Profiling

For each feature:

- Observe overall **data distribution**
 - Consider **value ordering** (rank behavior)
 - Check how values are spread (dense, skewed, or repetitive)
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Feature Grouping (Conceptual Only)

Group features based on:

- **Distribution shape** (e.g., smooth vs skewed)
- **Rank behavior** (well-ordered vs many ties / low resolution)

The goal:

Identify how each feature behaves statistically, not just its type (continuous/discrete)

No fixed rules here — this step is **data-driven and flexible**, and can be refined later.

Scaling Strategy

Assign a suitable normalization method for each group:

- Map all values into a **common range** [0, 1]
 - Use flexible scalers (e.g., linear or piecewise)
 - Ensure transformation preserves **relative ordering**
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Discrete Handling

- Convert values into a **bounded integer range** (e.g., up to 20k–30k)
 - Apply rounding if needed
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2. Embedding

Value Embedding

- Map each normalized value (integer) → vector of dimension d

Feature Embedding

- Map each feature (column identity) $\rightarrow$ vector of dimension k
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Combine

- Concatenate:

$$v_{\text{out}} = v_{\text{value}} \parallel v_{\text{feature}}$$

3. Row Representation

- Each feature $\rightarrow$ one vector
 - One row $\rightarrow$ a matrix (sequence of feature tokens)
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4. Model Input

- Feed the matrix into a deep learning model (e.g., attention-based)
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One-Line Idea

Treat each feature value as a **token informed by its distribution and rank behavior**, then learn interactions via embeddings.